

Exploring Spatio-Temporal Carbon Emission Across Passenger Car Trajectory Data

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Abstract—Carbon emissions caused by passenger cars in cities are essentially responsible for severe climate change and serious environmental problems. Exploring carbon emissions from passenger cars helps to control urban pollution and achieve urban sustainability. However, it is a challenging task to foresee the spatio-temporal distribution of carbon emission from passenger cars, as the following technical issues remain. *i)* Vehicle carbon emissions contain complex spatial interactions and temporal dynamics. How to collaboratively integrate such spatial-temporal correlations for carbon emission prediction is not yet resolved. *ii)* Given the mobility of passenger cars, the hidden dependencies inherent in traffic density are not properly addressed in predicting carbon emissions from passenger cars. To tackle these issues, we propose a Collaborative Spatial-temporal Network (CSTNet) for implementing carbon emissions prediction by using passenger car trajectory data. Within the proposed method, we devote to extract collaborative properties that stem from a multi-view graph structure together with parallel input of carbon emission and traffic density. Then, we design a spatial-temporal convolutional block for both carbon emission and traffic density, which constitutes of temporal gate convolution, spatial convolution and temporal attention mechanism. Following that, an interaction layer between carbon emission and traffic density is proposed to handle their internal dependencies, and further model spatial relationships between the features. Besides, we identify several global factors and embed them for final prediction with a collaborative fusion. Experimental results on the real-world passenger car trajectory dataset demonstrate that the proposed method outperforms the baselines with a roughly 7%-11% improvement.

Index Terms—Passenger cars, carbon emissions, collaborative spatial-temporal prediction, urban region, trajectory data.

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I. INTRODUCTION

CARBON emissions from various urban vehicles represent approximately 90% of urban air pollution [1], [2]. In particular, carbon emissions from vehicles in cities are essentially responsible for severe climate change and serious environmental problems [3]. As a solution, exploring the urban carbon emissions helps to monitor the air pollution situation and pave the way for achieving emissions reduction and energy conservation.

It is noteworthy that passenger cars (i.e., private cars)¹ are becoming the biggest contributor to urban carbon emission among various transportation modes. Such assertion stems from the fact that passenger cars are the main body participating in urban travels [4], [5]. According to [6], nearly 88.6% of the urban vehicles (223 million) are passenger cars in China (More than 76.6% in the European Union [7]). Although electric and hybrid cars appeared in passenger cars, and the proportion of these cars in newly sold cars is already very high, the total proportion of new energy cars is not high due to the large basic ownership of fuel-powered passenger cars. Among China's motor vehicles, new energy cars only account for 4%. Based on this data, it is believed that the electrification of passenger cars has not become common. In comparison, taxis and public transportation (e.g., bus and subway) have vigorously prompted electrification and new energy, generating few parts of urban carbon emissions.

In this line, massive carbon emissions from passenger cars have caused serious urban environmental problems, bringing great pressure on attempts to meet urban emission reduction targets [8]. To resolve this issue, exploring passenger car carbon emissions is worth delving into. To specify, it is required fine-drawn monitoring and forecasting the spatio-temporal carbon emission distribution from passenger cars to control urban traffic pollution and achieve carbon neutrality. On top of that, carbon emission prediction from passenger cars helps transportation managers and policymakers to improve the design of the city's transportation infrastructure [9].

To explore the spatio-temporal carbon emissions, existing efforts, including model-based and data-driven methods, have been proposed. The model-based methods are founded on an emission factor model, which rely on several relevant variables

¹In this paper, there is no distinction between the term 'passenger cars' and the term 'private cars'. Both refer to as vehicles with seven seats or less, and are non-operational, non-commercial and non-official cars. In this sense these two terms are interchangeable.

as input and output spatial-temporal predictions. On this basis, the microscopic emission factor models have been widely used in many countries and institutions [10], [11]. However, those methods are unsuitable for large-scale prediction since they depend highly on pre-defined knowledge and parameter settings. Data-driven methods consider emission prediction as a time series analysis problem analogous to traffic prediction, most of which exploit the spatial and temporal characteristics of vehicle emission to infer the future emission [12], [13]. For instance, Xu et al. introduced a convolutional neural network (CNN) architecture to achieve region-scale emission prediction [14] while neglecting the intrinsic spatial dependencies in the road network. Zhang et al. proposed a long short-term memory (LSTM) network and explored the time correlation of vehicle emission for estimating the future emissions [15]. This method suffers from gradient vanishing and a short memory span. Xu et al. [16] utilized emission sensor data and designed a spatial-temporal multifusion network for urban vehicle emission prediction. This method lacks generalizability and ignores the impact of geographic and semantic factors subjected to a fixed sensor device. However, these models are not suitable for large-scale prediction tasks. Spatio-Temporal Graph Convolutional Network (STGCN) [17], Optimized Graph Convolution Recurrent Neural Network (OGCRNN) [18] and Graph Wavenet (GWNEN) [19] achieved better learning latent representation on graph.

Motivated by the above-mentioned works, we strive to extract the spatial and temporal dependencies of carbon emission by utilizing trajectory data from passenger cars. It is expected to provide useful insights into the spatial distribution and temporal variation of carbon emissions. Although such an idea is intriguing, exploring the spatio-temporal distribution of carbon emission faces the following challenges:

- *Complex spatial-temporal correlations in carbon emission.* On the spatial scale, carbon emission in each region is greatly correlated with its adjacent neighbors, which is affected by the diffusion of carbon emission. On a temporal scale, carbon emissions change dynamically over time. For example, the traffic congestion occurring at 10 am will affect that of 11 am, causing an increase in carbon emissions in that region, of which the close time slots are more correlated than the distant. Besides, carbon emission shows periodic change patterns since weekday traffic congestion is usually similar. How to collaboratively integrate the spatio-temporal correlations for carbon emission prediction is challenging.
- *Hidden dependencies underlying traffic density.* Different from general time series forecasting problems, the evolving trend of carbon emission has internal dependencies, i.e., related traffic features such as traffic density exhibit synchronous change patterns, providing auxiliary clues to facilitate carbon emission forecasting. For instance, the future carbon emission cannot only be correlated to the current observations but also be impacted by traffic density [16]. However, most existing methods ignore the hidden dependencies inherent in traffic density when predicting carbon emissions. Exploring the potential

relationship between carbon emissions and traffic density is a challenge.

To address the above-mentioned challenges, we propose a Collaborative Spatial-temporal Network **CSTNet**, where we model spatial heterogeneity from the multi-view graph, associated with multi-variable input of primary and auxiliary feature to collaboratively predict carbon emissions from passenger cars. Specifically, we first model the heterogeneous spatial correlations using graph structure with diverse perspectives, which are based on spatial connectivity, geographical proximity, and temporal similarity. Following that, a spatial-temporal convolutional (ST-Conv) block, including temporal gate convolution, spatial convolution, and temporal attention mechanism, is proposed to capture the spatial interactions and temporal dynamics of both primary and auxiliary features. Second, to handle the internal dependencies of traffic density on carbon emission, we design an interaction layer to realize the interaction between them and integrate their hidden states from a graph perspective. Additionally, we identify various global factors, including driving behaviors and external features, and then embed the factors into the output layer for final collaborative prediction. Last but not least, we visualize the results to intuitively interpret the distribution of urban carbon emissions.

The main contributions in this paper are summarized as follows.

- We propose a collaborative spatial-temporal network (CSTNet) for carbon emission prediction. We model the heterogeneous spatial correlations using graph structure with multiple perspectives and propose a spatial-temporal convolutional block to capture the spatial interactions and temporal dynamics. In this way, we implement collaborative prediction by a parallel input of carbon emission and traffic density.
- We design an interaction layer to handle the internal dependencies of traffic density on carbon emissions and integrate their hidden states. Via designing a fusion strategy for collaborative prediction, the spatial-temporal representations are further combined with global factors such as driving behaviors and external features.
- There is good interaction between the main and auxiliary features, which share the same convolutional structure. These two features can be effectively combined with global factors through an adaptive fusion strategy.

The remainder of this paper is organized as follows. Sec. II presents the related works. In Sec. III, we introduce notations and preliminaries used in this paper, and then we formalize the predicting problem of carbon emission. The details of the proposed model are presented in Sec. IV. After that, Sec. V presents experimental results based on the real-world trajectory dataset. Finally, we conclude the paper in Sec. VI.

II. RELATED WORKS

There are two main classes of approaches for predicting vehicle carbon emission: model-based and data-driven. The model-based methods utilize emission factor models, in which

the vehicle trajectory and other relevant variables are input and output a spatial-temporal prediction. The widely used emission factor models developed in the USA and Europe estimate the total carbon emissions in a region for a given time period. These methods include a computer program to calculate emissions from road transport (COPERT) [10], motor vehicle emission simulator (MOVES) [20] and international vehicle emissions (IVE) [11]. Following that, the GWR model was employed by Zou et al. [21] to predict urban pollution emissions, which is sensitive to environmental factors, such as humidity, temperature, pressure, and geographical locations. However, these models are not suitable for large-scale prediction tasks since they are based on prior knowledge, and their performances are highly dependent on parameter settings.

Guo et al. [22] proposed an Attention-based Spatial-Temporal Graph Neural Network (ASTGNN) for traffic forecasting. Li et al. [23] proposed Spatial-Temporal Fusion Graph Neural Networks (STFGNN) for traffic flow forecasting. Zhang et al. [24] proposed Crossformer, a Transformer-based model utilizing cross-dimension dependency for MTS forecasting. Liu et al. [25] reflect on the competent duties of Transformer components and repurpose the Transformer architecture without modifying the basic components. Nie et al. [26] proposed an efficient design of Transformer-based models for multivariate time series forecasting and self-supervised representation learning.

In [5], the authors provided a new perspective on human mobility, focusing on passenger cars, by investigating mobility patterns and capturing the spatiotemporal evolution of urban hot zones from people's arrive-stay-leave (ASL) behavior. Xiao et al. [4] proposed a deep learning framework for a spatiotemporal attention network (STANet) with a neural algorithm logic unit (NALU), the so-called STANet-NALU, to understand the dynamic aggregation effect of passenger cars on weekends. [27] explored intercity mobility in urban agglomerations by surveying people traveling across cities based on private car trajectory data. The authors in [28] established a deep learning model based on long short-term memory (LSTM) to capture the evolution of the area attractiveness. In [29], the authors proposed a classification method based on transfer learning to migrate existing knowledge to solve learning problems in the target domain with only a small amount of labeled trajectory data or even no data. Huang et al. [30] focused on investigating the individual travel patterns of private car users based on a large-scale private car trajectory dataset. Wang et al. [31] investigated the SAW behavior and aggregation effect based on large-scale private car trajectory data. Xiao et al. [32] proposed a Hierarchical Spatial-Temporal Network (HierSTNet) to forecast urban region heat, which contains two representations, namely, grid region from micro perspective and node region from macro perspective.

Xiao et al. [33] proposed a low-cost and user-friendly implementation method for achieving large-scale private cars trajectory data acquisition via designing a lightweight GPS module and On Board Diagnostics (OBD) reader. Previously, their team had done a lot of work [34], [35] to lay the technical groundwork for collecting private car trajectory

data. Havyarimana et al. proposed a reliable fusion technique, called non-Gaussian Redheffer weighted least squares (n GRWLSs) [33], and a novel collaborative vehicular localization approach termed non-Gaussian weighted least-squares (n GWLS) [36] for accurate intervehicle positioning. In [37], the author proposed a novel ensemble transfer regression framework for urban environments with transfer learning as the primary solution for constructing a fine-grained trajectory dataset during GNSS outages.

In the data-driven methods, the carbon emissions is regarded as a type of time series analysis similar to traffic prediction problem [15], [38], [39]. Azeez et al. [12] utilized support vector regression (SVR) to predict section-scale vehicle emission but with limited precision. Then, machine learning was proven to be suitable for vehicle emission prediction. For instance, Artificial Neural Networks (ANNs) were applied in prediction tasks to learn patterns from the high dimensional data. However, the hand-crafted features in these models are often shallow in structure and perform poorly. Zhang et al. [15] proposed a long short-term memory network (LSTM) and exploited the time correlation of vehicle emission on the same road section to estimate emission of the next time. Xu et al. [14] introduced a convolutional neural network (CNN) framework into vehicle emission prediction, which proved its effectiveness and achieved region-scale emission prediction. However, these methods ignore the successive spatiotemporal properties related to traffic conditions and fail to model complex spatial topology structures.

At recent stage, spatial-temporal graph neural networks (GNN) have been widely applied to the profiling of traffic-related problems since graph representation is a natural way to express the intrinsic structure of the road network. For example, Yu et al. [17] proposed a convolutional structure for traffic forecasting. Guo et al. [18] designed a deep learning framework called OGCRNN, combining RNN and the residual graph to replace the empirical and fixed graph structure. Wu et al. [19] added an adaptive term by generating one learnable correlation matrix to extend diffusion convolution for better learning latent representation on the graph. However, these researches have not been able to fully utilize the spatial topological information among urban roads, generally ignoring more semantic factors. Following that, the researchers aimed to effectively capture more semantic correlations and model the spatial heterogeneity. Later, these approaches utilized emission sensor data and multi-source urban data to consider the intrinsic structure of the road network and traffic conditions. Elmi et al. [13] designed a deep-learning-based model to forecast energy consumption in each and every road in cities based on real traffic conditions. Xu et al. [16] designed a spatial-temporal multifusion network and extracted the spatial-temporal dependencies together with external factors for urban vehicle emission prediction. However, these methods suffer from poor numerical extrapolation, and some geographic and semantic factors might not be available to fix sensor devices.

Unlike the existing methods, we propose a collaborative spatial-temporal network (CSTNet) for carbon emission prediction from passenger cars, which contains a multi-view

TABLE I
TRIP INFORMATION IN PASSENGER CAR TRAJECTORY DATA

UserID	StratTime	StartLat	StartLon	StopTime	StopLon	StopLat	TravelMile	TravelTime	TravelOil	Speed	SpeedVariation
556912	2020/10/1 13:26	113.902853	22.563991	2020/10/1 14:13	113.803697	22.718948	22014	2802	1.37	16	0
556912	2020/10/1 18:25	113.822951	22.69896	2020/10/1 18:49	113.866116	22.572824	15923	1448	2.67	40	1
556912	2020/10/1 20:30	113.866249	22.573196	2020/10/1 21:57	113.822222	22.699113	16322	5201	2.71	38	1

graph structure with a parallel input of carbon emission and the auxiliary feature for collaborative prediction. Besides, we design an interaction layer to represent the internal dependencies between carbon emissions and auxiliary features. Various global factors, including driving behaviors and external features, are embedded into the output layer for final collaborative prediction.

III. DEFINITIONS AND PROBLEM STATEMENT

A. Definitions

Definition 1 (Carbon Emission From Private Cars): We retrieve trip information from the passenger car trajectory data (as depicted in Table I), which records trip time, locations, driving status and fuel consumption. After obtaining the fuel consumption of each region i from trip information of passenger car trajectory data, the corresponding carbon emissions can be calculated based on emission factors [2], [40]:

$$\mathcal{R}_i = \sum_k \mathcal{O}_{i,k} \times e, \quad (1)$$

where $\mathcal{O}_{i,k}$ denotes the actual fuel consumption in each region i of passenger car k (unit: liter, see 'TravelOil' in Table I. Refer to V-A for more details). e is the emission factor of CO₂, which is set to 2.18kg/L according to the Intergovernmental Panel on Climate Change (IPCC) report [40]. $\mathcal{R}_i$ is the corresponding carbon emissions of region i (unit: kg). Considering the difference between the collected passenger cars and the actual number of urban vehicles in a specified region, we use the average estimation method to model each regional carbon emission. Suppose $\mathcal{P}_i$ is the number of the collected passenger cars in region i , the carbon emissions can be estimated by:

$$\mathcal{V}_i^u = \frac{\mathcal{R}_i}{\mathcal{P}_i}. \quad (2)$$

We define the carbon emissions of region i at time slot t by $\mathcal{V}_{i,t}^u \in \mathbb{R}$. Then, $\mathcal{X}_t^u = \{\mathcal{V}_{1,t}^u, \dots, \mathcal{V}_{N,t}^u\} \in \mathbb{R}^N$ can represent all carbon emission observations at the t -th time interval, where N is the number of regions.

Definition 2 (Auxiliary Feature): Underlying urban emission networks with dynamic traffic, there exists a hidden correlation between traffic volume and carbon emission, which can furnish auxiliary clues to facilitate carbon emission prediction. As an average representation of traffic flow, we take traffic density as an auxiliary feature to predict carbon emissions collaboratively. Suppose the area of region i is $s_i \times s_i$ and the traffic density can be expressed by:

$$\mathcal{V}_i^a = \frac{\mathcal{P}_i}{s_i}. \quad (3)$$

We use $\mathcal{V}_{i,t}^a \in \mathbb{R}$ to represent the auxiliary feature of i -th region at time slot t . Then, the set of regions with

historical observations of auxiliary features is denoted as $\mathcal{X}_t^a = \{\mathcal{V}_{1,t}^a, \dots, \mathcal{V}_{N,t}^a\} \in \mathbb{R}^N$.

B. Problem Statement

Carbon emission prediction aims to predict the average carbon emission for each region in the city historical observation in a period of future time. Given a historical window period p of carbon emissions and carbon density, our goal is to learn a model f accompanied with global feature $\mathcal{X}^e$ to collaboratively predict future q time slots of urban carbon emissions,

$$\hat{\mathcal{X}}_{t+1:t+q} = f(\mathcal{X}_{t-p+1:t}^u, \mathcal{X}_{t-p+1:t}^a, \mathcal{X}^e). \quad (4)$$

IV. METHODOLOGY

In this section, we present the design of the proposed CSTNet in detail. The overall framework of CSTNet is shown in Fig. 1. To specify, we design the urban region construction to divide the urban area into discrete grids. The multi-view graph modeling module generates multiple collaborative graphs with spatial and semantic correlations. The proposed CSTNet includes two parallel parts: one for the primary feature and one for the auxiliary feature. The primary part consists of a spatial-temporal convolutional block stacking L layers, which is used to model the primary feature, i.e., the carbon emissions $\mathcal{X}^u$. The auxiliary part is used to model the auxiliary feature, namely the traffic density $\mathcal{X}^a$. To fuse the features, an interaction layer takes an adaptive learning block to realize the interaction with the primary feature and integrate their hidden states for input to the next layer. The outputs of the primary part in each layer are passed through skip-connection and integrated into the output layer, embedding global features for collaborative prediction of carbon emissions. Lastly, we visualize the results to intuitively interpret the distribution of urban carbon emissions.

A. Urban Region Construction

1) *Pre-Processing:* As shown in Table I, the trajectory data of passenger cars are stored in individual trips, containing location information (i.e., trip time, latitude, and longitude) and driving status of the vehicle (i.e., fuel consumption and average speed). The details can be seen in Sec. V-A. We extract the information of fuel consumption from the selected trajectory dataset to estimate carbon emission (see Sec. I). Referring to the analysis of [41], the driving behaviors of speed variation is the key parameter in vehicle emissions estimation. Hence, we take the average speed (see 'Speed' in Table I) and count the number of rapid acceleration and deceleration (see 'SpeedVariation' in Table I) as global information from trajectory data to facilitate carbon emission prediction.

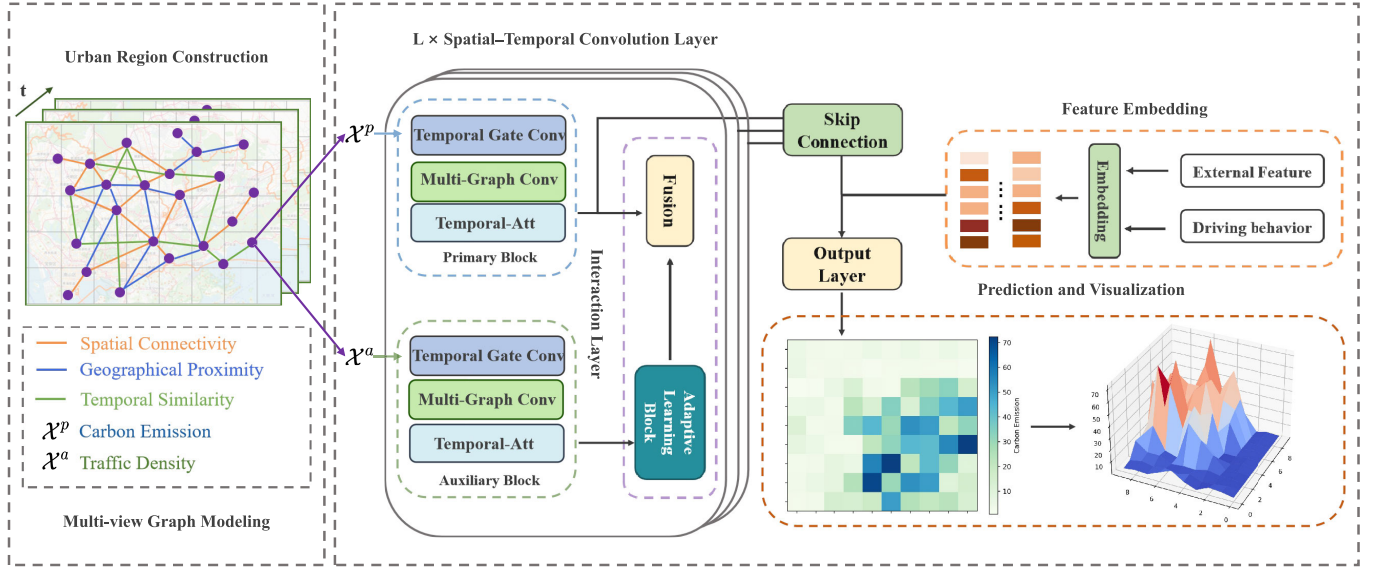


Fig. 1. The proposed CSTNet for carbon emission prediction.

Note that 20% of the total emissions from transportation are cold start emissions [42], indicating that distinguishing the cold-start phase and hot phase in vehicles' trajectories is beneficial for our prediction task. To that end, we encode the status of cold-start by retrieving trajectory data. In detail, for analyzing each trip in trajectory, it can be interpreted that the user's arrival matches the stop (off the engine) in the arrive-stay-leave behavior [43]. The leave denotes the start point of the next trip. We define the cold start as a car that takes more than two hours from the last stop to the next start. For example, in Table I, the user arrived at a specified place at 14:13 then left at 18:25 and stayed more than two hours, thus we encode it as 1. Likewise, the next stay due to the engine off and on again lasts less than two hours, which is encoded as 0. Besides, we design a data cleaning algorithm to remove the trajectory data supplemented with time and speed. We use the relative distance at the previous moment as the threshold. A distance of less than 10 meters will be regarded as an error value to screen out, as such data will not appear during actual driving processes.

2) *Region Modeling*: Discretizing continuous space helps to process and analyze spatial data. The gridding technique is one of the discretization methods, and this technique is widely used in models such as layered time series, grouped time series, and ARIMA [44], achieving good results. Considering the free diffusion of carbon dioxide gas in space, using a two-dimensional grid as the prediction unit is more suitable than a one-dimensional road segment. As for the start and stop location, we employ a gridding technique to divide the geographic space by combining all trajectories into discrete grids. Specifically, given grid size s , the city map is equally divided into N grids. Each of the grids is considered a small urban region, and we design a region mapping function so that the trajectory data can be mapped up. Specifically, it assigns a journey segment to the grid where its stop point is located. Through this allocation, we can easily obtain the number of

trips in the grids, which is also the number of passenger cars in a short period of time (if time is extended, data accuracy can also be ensured by deduplicating based on 'UserId'). By doing so, we can understand the spatio-temporal clustering of passenger cars. Besides, using the gridding technique eliminates the need to consider the relationship between coordinate data and real locations, greatly reducing workload. Through such grid division and selecting different time periods, we can easily analyze the temporal and spatial aggregation levels.

B. Multi-View Graph Modeling

After dividing the city into a grid map, we convert the time series data of carbon emission and traffic density into graph structure frames. In this section, considering semantic information about passenger cars' travel and spatial contexts, we introduce three different views based on constructed graphs. More generally, each region, i.e., the vertexes in a graph structure, connect to multiple neighbors via different views, of which the supplementary provides rich spatial and semantic information for prediction tasks. For each type of spatial relationship $r \in \mathcal{R}_{[c,d,t]}$, we build the adjacent matrix $\mathcal{A}_r = (a_{ij})_{N \times N}$ for each graph $\mathcal{G}_r$.

1) *Spatial Connectivity-Based Graph*: The topological structure of each region can be represented as a graph $\mathcal{G}_c = (\mathcal{V}, \mathcal{E}_c, \mathcal{W}_c)$, where $\mathcal{V}$ is the set of nodes. The weight $\mathcal{E}_c(i, j)$ of edges $\mathcal{E}_c$ indicates the connectivity between two nodes. That is, an edge is created if a trajectory starts from region i/j and stays at region j/i . The mathematical representation of the connectivity graph is denoted by the adjacent matrix $\mathcal{A}_c \in \mathbb{R}^{N \times N}$.

2) *Geographical Proximity-Based Graph*: Carbon emissions in each region can impact that of its spatial neighbors. For example, the future carbon emissions in the downstream are affected by the emission diffusion of nearby regions, and it is intuitive that the adjacent neighbors are more relevant than the distant ones. To capture the correlations among

TABLE II
MAIN NOTATIONS AND DEFINITIONS

Notations	Definitions
s_i	The grid size of the region i
r	Types of spatial relationships, which is represented as a graph
c	Spatial connectivity
d	Geographical proximity
t	Time similarity
R_i	The calculation of carbon emission in the region i
$O_{i,k}$	The fuel consumption in each region i of passenger car k
P_i	The amount of passenger cars in the region i
$\mathcal{V}_i^u$	The average calculation of carbon emission in the region i
$\mathcal{X}_t^u$	The all carbon emission observations, as the primary feature at the t -th time intervals
$\mathcal{V}_i^a$	The calculation of traffic density in the region i
$\mathcal{X}_t^a$	The auxiliary feature representation at the t -th time windows
$\tilde{\mathcal{X}}$	The output representation in temporal gate convolution, including primary feature $\tilde{\mathcal{X}}^u$ and auxiliary feature $\tilde{\mathcal{X}}^a$
$\mathcal{H}_r$	The output representation in spatial convolution block, including $\mathcal{H}_r^u$ and $\mathcal{H}_r^a$
E	The normalized matrix of temporal correlation
$\hat{\mathcal{H}}$	The output representation in temporal attention block, including $\hat{\mathcal{H}}^u$ and $\hat{\mathcal{H}}^a$
$\mathcal{A}_{adp}$	The adaptive learning matrix
$\hat{\mathcal{H}}_{ap}$	The effect of auxiliary feature on primary feature
$\mathcal{H}_p$	The aggregated representation of primary feature
$\mathcal{H}_{st}$	The final representation of primary feature

spatial locations, we construct a geographical proximity-based graph $\mathcal{G}_d = (\mathcal{V}, \mathcal{E}_d, \mathcal{W}_d)$ to explore the strength of the impact in terms of actual physical distance:

$$\mathcal{A}_d(i, j) = \begin{cases} 1, & \text{Haversine}(i_c, j_c) < d \text{ and } i \neq j \\ 0, & \text{others,} \end{cases} \quad (5)$$

where the calculation of weights between two linked nodes i and j is based on Haversine distances, d is the average Haversine distance of all pairs of regions in the map. We find the coordinates of the center of mass for each region using iterative K-mean aggregation of historical data, namely i_c and j_c . Nodes that are closer will be correlated with higher weights.

3) *Time Similarity-Based Graph*: As affected by traffic conditions, the carbon emissions of urban regions change over time. Such temporal trends will be similar despite the fact that the two regions are geographically distant or spatially disconnected. With the historical observations data, given time sequence tl_i, tl_j showing variations of two regions in the temporal scale, we employ a DTW (Dynamic Time Warping) function to calculate the distance between tl_i and tl_j , represented as $DTW(tl_i, tl_j)$. Then, we transform it to a similarity measurement and construct a time similarity-based graph $\mathcal{G}_t = (\mathcal{V}, \mathcal{E}_t, \mathcal{W}_t)$ to reveal the similarity of carbon emission trends:

$$\mathcal{A}_t(i, j) = \exp\left(-\frac{DTW(tl_i, tl_j)}{\varepsilon^2}\right), \quad (6)$$

where ε is the parameter of distance attenuation.

C. Spatial-Temporal Convolution

Following the constructed graph with multiple perspectives, we design a spatial-temporal convolutional block to capture spatial heterogeneity by convolution operation, which can effectively capture local dependencies and maintain shift invariance in both spatial and temporal dimensions.

1) *Temporal Gate Convolution*: Carbon emissions for one region are highly associated with historical observations. Motivated by this, we strive to capture the correlation in time dimension by consolidating dilated causal convolutions and gated linear units (GLU) [45] in L layers. Compared to the RNN-based approach, dilated convolution is favorable for saving computational consumptions and avoid gradient explosion when the information across time spans is harvested. The dilated causal convolution of filter $\mathbf{f} \in \mathbb{R}^K$ at t step can be expressed by:

$$\mathcal{X}_t^u \star \mathbf{f}(t) = \sum_{s=0}^{K-1} \mathbf{f}(s) \mathcal{X}_{t-d \times s}^u, \quad (7)$$

where d is the dilation rate to control skipping distance. As the depth of layers increases, dilated convolution receives wider receptive fields. Thanks to the mighty controlling information flow through layers [45], we employ GLU structure to parcel the previous convolution operations:

$$\tilde{\mathcal{X}}^u = GLU(\mathcal{X}_t^u) = g(\Theta_1 \star \mathcal{X}_t^u + \mathbf{b}_1) \odot \sigma(\Theta_2 \star \mathcal{X}_t^u + \mathbf{b}_2), \quad (8)$$

where Θ_1, Θ_2 are a group of kernels, $\mathbf{b}_1$ and $\mathbf{b}_2$ are model biases, $\odot$ is the element-wise product, g and σ are both activation functions.

2) *Spatial Convolution*: The graph convolutional network (GCN) provides a promising way to aggregate spatial features of nodes given pre-defined graph structure [46]. As shown in Fig. 2, we encode the spatial heterogeneity with multiple perspectives in a graph structure, and each graph is input to an individual GCN. The information propagation from neighbor nodes of GCN is formulated by:

$$\mathcal{H}_p^{(l+1)} = \left(\tilde{\mathcal{D}}^{-\frac{1}{2}} \tilde{\mathcal{A}}_r \tilde{\mathcal{D}}^{-\frac{1}{2}} \mathcal{H}_p^{(l)} \mathcal{W}_r^{(l)}\right), \quad (9)$$

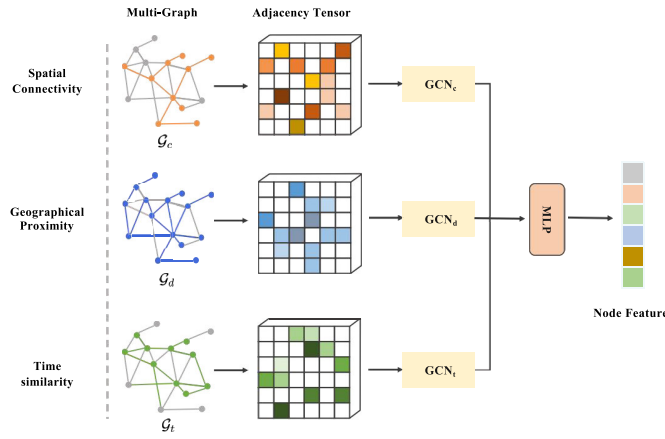


Fig. 2. The architecture of spatial convolution module.

$$\tilde{\mathcal{A}}_r = \mathcal{A}_r + I_N, r \in \mathcal{R}_{[c,d,t]}, \quad (10)$$

where $\tilde{\mathcal{A}}_r$ is the adjacent tensor adding self-connections, I_N is an identity matrix, $\mathcal{W}_r^{(l)}$ is the layer-wise trainable weight, $\tilde{\mathcal{D}}$ is the diagonal matrix and $\tilde{\mathcal{D}}[i, i] = \sum_j \tilde{\mathcal{A}}[i, j]$. The propagation can be considered as a spectral filter in the Fourier domain. Each node region is input to multiple GCN_r (i.e., GCN_c , GCN_d and GCN_t) together with the corresponding graph $\mathcal{G}_r$ (i.e., $\mathcal{G}_c$, $\mathcal{G}_d$ and $\mathcal{G}_t$), resulting in the output feature matrices $\mathcal{H}_r$ (i.e., $\mathcal{H}_c$, $\mathcal{H}_d$ and $\mathcal{H}_t$) for each layer. We utilize an MLP layer to fuse the feature matrices $\mathcal{H}_r$:

$$\tilde{\mathcal{H}} = \text{softmax}(\sum_r \mathcal{W}_r \circ \mathcal{H}_r), \quad (11)$$

where $\circ$ denotes the element-wise product. We conduct the softmax operation to normalize the feature matrix.

3) *Temporal Attention Mechanism*: For time series data with intricate temporal dependencies, solely relying on Temporal Gate Convolution might not be sufficient to capture all underlying patterns. Temporal Gate uses convolutional kernels to model local patterns and trends but lacks global learning for extracted features. In this line, we leverage a time-wise attention mechanism [47] to dynamically focus on crucial time steps, providing more valuable information.

Let E denote the normalized matrix of temporal correlation, which can be formulated by,

$$E = (V_e \cdot \sigma(((\mathcal{H}^{(l-1)})^T U_1) U_2 (U_3 \mathcal{H}^{(l-1)}) + b_e)), \quad (12)$$

$$E' = \text{softmax}(E), \quad (13)$$

where V_e, U_1, U_2, U_3, b_e are learnable parameters. $E_{i,j}$ in E denotes the strength of dependencies between time i and j . The input is tuned by the attention matrix. Then, we can obtain $\hat{\mathcal{H}}_p = \mathcal{H}_p E'$ to informatively adjust the input in temporal dimension.

Besides the primary part, including temporal gate convolution, spatial convolution and temporal attention mechanism, we design a similar structure of spatial-temporal convolutional block for the auxiliary part. The auxiliary feature of traffic density shares the same graph structure as carbon emission, and the output is denoted by $\hat{\mathcal{H}}_a$, accordingly. In the following section, we will describe the interaction layer between the primary and auxiliary features in detail.

D. Interaction Layer Between Primary and Auxiliary Features

For handling the internal dependencies of traffic density on carbon emission, we propose an interaction layer to realize the interaction between the primary feature and auxiliary feature and merge their hidden states. The interaction layer consists of an adaptive learning block with a fusion strategy.

1) *Adaptive Learning Block*: In this section, we propose an adaptive learning matrix $\mathcal{A}_{adp}$ to discover the implicit correlations between carbon emission and traffic density. To specify, we introduce two node embedding dictionaries with learnable parameters to derive the spatial dependency weights adaptively. However, considering the internal dependencies might change due to the dynamic properties of the real traffic density, we envision adding a time dimension for the correlation matrix to represent the dynamic relationships [48]. Accordingly, the adaptive learning adjacency matrix $\mathcal{A}_{adp} \in \mathbb{R}^{N_p \times N_a \times N_t}$ is constructed by matrix multiplication of three dimensions:

$$\mathcal{A}_{t,i,j} = \sum_{k=1}^d \sum_{i=1}^d \sum_{j=1}^d \mathbf{E}^c \mathbf{E}_i^s \mathbf{E}_j^t, \quad (14)$$

$$\mathcal{A}_{adp} = \text{softmax}(\text{RELU}(\mathcal{A}_{t,i,j})), \quad (15)$$

where we define $\mathbf{E}^s \in \mathbb{R}^{N_s \times d}$ as source nodes embedding, $\mathbf{E}^e \in \mathbb{R}^{N_e \times d}$ as target nodes embedding, $\mathbf{E}^t \in \mathbb{R}^{N_t \times d}$ as time slots embedding. We assign a learnable core tensor $\mathbf{E}^c \in \mathbb{R}^{d \times d \times d}$. Accordingly, N_t, N_s, N_e, d denotes the number of time slots, the number of source nodes, the number of target nodes, and the number of embedding dimensions, respectively. Since we attempt to predict the future average carbon emissions accumulated on a region scale, the corresponding time slot N_t is set to 24, which means an hourly collection.

2) *Fusion Layer*: After the parallel structure of spatial-temporal convolutional block for the primary feature and auxiliary feature in each layer, we model their inter-dependencies and integrate the hidden states of auxiliary feature $\hat{\mathcal{H}}_a$ with the primary feature $\hat{\mathcal{H}}_p$:

$$\hat{\mathcal{H}}_{ap}^{(l)} = \mathcal{A}_{adp} \hat{\mathcal{H}}_a \mathcal{W}_{ap}, \quad (16)$$

$$\tilde{\mathcal{H}}_p^{(l)} = \text{Concat}(\hat{\mathcal{H}}_p^{(l)}, \hat{\mathcal{H}}_{ap}^{(l)}), \quad (17)$$

$$\tilde{\mathcal{H}}_p^{(l)} = \tilde{\mathcal{H}}_p^{(l)} \mathcal{W}_1 + b_1, \quad (18)$$

where $\hat{\mathcal{H}}_{ap}^{(l)}$ represents the effect of auxiliary feature on primary feature. We learn their inter-dependencies by conducting convolution operations on an adaptive graph $\mathcal{A}_{adp}$. Here $\mathcal{W}_{ap}, \mathcal{W}_1$ and b_1 are learnable parameters. Then, we choose the concatenate function to aggregate this hidden information. The integrated states $\tilde{\mathcal{H}}_p^{(l)}$ are fed into a fully connected layer, and the output in each layer (primary part and auxiliary part) are passing through the next layer as input. Besides, to extract more characteristics from the different stages, we add a skip connection for each output of the primary part to the output layer. The final states of primary feature are passed a MLP to perform linear transformation. Then we obtain

$$\tilde{\mathcal{H}}_{st} = \|\|_{l=1}^L \tilde{\mathcal{H}}_p^{(l)}[:, -l : :], \quad (19)$$

$$\mathcal{H}_{st} = \text{Relu}((\tilde{\mathcal{H}}_{st} \mathcal{W}_2 + b_2) \mathcal{W}_3 + b_3), \quad (20)$$

where $\parallel$ is the concatenating operation of L layers, $\mathcal{W}_2, \mathcal{W}_3, b_2$ and b_3 are parameters.

E. Feature Embedding

In addition to spatial-temporal dependencies, carbon emissions are affected by driving behaviors (i.e., speed variation and cold start) and other external factors (i.e., weather condition) [41], [42]. In this subsection, we extract the global features and embed them into the output layer.

1) *Driving Behaviors*: The findings in [42] reveal that 20% of the total emissions from vehicles are cold start emissions. Given the passenger car trajectory data, we need to distinguish the variation of emission in the cold-start phase and hot phase, respectively. Thus, We encode the status of cold start and denote it as E_{CS} . The speed variation is considered the key parameter for estimating vehicle emissions [41]. As such, we take the average speed and the count of rapid speed variation (see 'Speed' and 'SpeedVariation' in Table I) for each trip into the modeling of carbon emission prediction. Specifically, we extract the average speed feature at future time $t+q$ and encode the count of sharp acceleration and deceleration, which is represented as SF_t and E_{AD} , respectively.

2) *External Factors*: External factors such as weather conditions provide additional information for predicting carbon emissions. In light of that, we take it into the feature embedding, and the attributes of weather are divided into several categories, i.e., sunny, rainy, foggy, and windy, denoted as E_{WC} . Besides, for all introduced features of discrete values that cannot be fed directly to the neural network (such as E_{CS} , E_{AD} and E_{WC}), we adopt one-hot encoding method to convert these data into low-dimensional vectors.

3) *Embedding Layer*: We stack two fully connected layers to integrate various features with different dimensions. First, we fuse SF_t and discrete feature into a long feature tensor $\mathcal{X}_e = E_{CS} \oplus E_{AD} \oplus E_{WC} \oplus SF_t$, where the operation $\oplus$ means the addition of a matrix and a vector. Then, the second layer is used to ensure that the output tensor $\mathcal{X}_e$ matches the same shape with hidden states of carbon emissions $\mathcal{H}_{st}$. The sum fusion $\mathcal{X}_e$ is fed into the output layer for final prediction. The predicted value in the next q time slices is denoted by $\hat{\mathcal{X}}_{t+1:t+q}$. Finally, our model is trained to minimize the loss of Mean Absolute Error (MAE) between predictive carbon emissions and true values, formally:

$$\hat{\mathcal{X}}_{t+1:t+q} = \tanh(\mathcal{W}_{st} \odot \mathcal{H}_{st} + \mathcal{W}_e \odot \mathcal{X}_e), \quad (21)$$

$$\mathcal{Loss} = \frac{1}{q \times N} \sum_{i=t+1}^{t+q} \sum_{j=1}^N \left| \mathcal{X}_{i,j}^u - \hat{\mathcal{X}}_{i,j}^u \right|. \quad (22)$$

V. EXPERIMENTS

A. Passenger Car Trajectory Datasets

We evaluate the carbon emission prediction performance of the proposed model based on real-world passenger car trajectory datasets. The passenger car trajectory data are collected by an integrated OBD/GPS device [49], [50]. The On-Board Diagnostics (OBD) unit records the driving status of the vehicle, and the GPS receiver collects location information, including trip time, latitude, and longitude. The running

TABLE III
PASSENGER CAR TRIP DATASETS IN CHANGSHA AND SHENZHEN

Cities	Changsha	Shenzhen
Latitude	27.92-28.21	22.51-22.83
Longitude	112.97-113.18	113.75-113.99
Time span	9/30/2020-1/1/2021	9/30/2020-1/1/2021
Number of trips	311,990	481,968
Number of regions	417	319

example of the passenger car trajectory dataset² contains over 5000 passenger cars and nearly 100,000 trip trajectories in an urban environment. Table I presents a sample of trip information in the passenger car trajectory data. The fuel consumption data is automatically collected by the on-board diagnostics (OBD) reader installed on private cars. The OBD reader is connected to the onboard computer and can read its fuel consumption. The "TravelOil" represents the fuel consumption during the period from starting to shutting down the car engine. Its specific collection and numerical calculation are performed by the equipment provider, who provided us with the generated dataset file for our research and use. We welcome researchers to directly use all the data in our dataset for research.

In the experiments, we select passenger car trajectory data collected from two metropolises in China, i.e., Changsha and Shenzhen, as illustrated in Table III. We retrieve three-month trajectory datasets from 30 Sep. 2020 to 1 Jan. 2021, which contains 311,990 and 481,968 trips, respectively. After discarding abnormal data, we adopt Z-score normalization to process the data. each dataset is split into 60% for training, 20% for validation, and 20% for test.

B. Baselines

We compare the proposed CSTNet with the following baselines, of which the parameters are fine-tuned. These methods include 2 statistic methods (HA, ARIMA), 3 RNN-based methods (LSTM, GRU, OGCRNN), 2 Transformer-based methods (iTransformer, Crossformer) and 2 GCN-based methods (STGCN, GWNET).

- **HA**: Historical Average is a classical approach using the average of historical data as the prediction output.
- **ARIMA** [44]: Autoregressive integrated moving average is a well-known model for understanding and predicting future values in a time series.
- **LSTM** [51]: Long Short-Term Memory Networks is a variant of RNN, which can capture the long-term temporal dependencies and avoid the vanishing gradient problem by gated regulators.
- **GRU** [52]: A recurrent neural network with fully connected GRU hidden units. Formally, GRU can be trained on sequences with the arbitrary length.
- **OGCRNN** [18]: A novel Optimized Graph Convolution Recurrent Neural Network (OGCRNN) for traffic prediction, in which it learns the spatial-temporal features by the latest variation of RNN and designs a residual graph model to replace the empirical and fixed graph.

²<https://github.com/HunanUniversityZhuXiao/PrivateCarTrajectoryData>

TABLE IV
EVALUATIONS OF DIFFERENT BASELINES

Dataset		Changsha				Shenzhen			
Method	Metric	30min		60min		30min		60min	
		MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
Statistic	HA	5.35	7.44	4.68	6.88	5.68	7.88	5.22	7.35
	ARIMA	3.89	6.05	3.00	4.72	4.00	5.72	3.87	5.23
	LSTM	2.43	4.51	1.78	3.09	3.18	4.69	2.33	3.86
RNN-Based	GRU	2.44	4.49	1.79	3.09	3.19	4.69	2.33	3.86
	OGCRNN	2.37	3.80	1.75	3.02	2.75	4.02	2.31	3.90
Transformer-Based	iTransformer	2.13	3.56	1.69	2.98	2.65	3.93	2.22	3.76
	Crossformer	2.20	3.42	1.68	2.96	2.70	3.89	2.23	3.69
	STGCN	2.24	3.69	1.71	2.91	2.71	3.91	2.25	3.76
GCN-Based	GWNET	2.09	3.46	1.69	2.84	2.69	3.84	2.30	3.74
	CSTNet	1.85	3.10	1.58	2.88	2.58	3.78	2.02	3.67

- **iTransformer** [25]: Changed the usage of transformer to make it work in predicting time series. This method embeds each time series as variate tokens, adopts the attention for multivariate correlations, and employs the feed-forward network for series representations.
- **Crossformer** [24]: Crossformer is a Transformer-based model utilizing cross-dimension dependency for MTS forecasting. Crossformer establishes a Hierarchical Encoder-Decoder (HED) to use the information at different scales for the final forecasting.
- **STGCN** [17]: This architecture comprises several spatio-temporal convolutional blocks, which is the first to apply a pure convolutional structure to simultaneously extract spatio-temporal features from graph-structured time series for the first time in a traffic study.
- **GWNET** [19]: A spatial-temporal graph convolutional network, which integrates diffusion graph convolutions with 2D dilated convolutions and develops a novel adaptive dependency matrix through node embedding.

In the proposed CSTNet model, we set grid size s to 2500 meters on both datasets, the according regions are divided into 417 in Changsha and 319 in Shenzhen. We set the number of GCN layers to 2, the number of ST-layer to 8 with a sequence of dilation factors [1, 2, 1, 2, 1, 2, 1, 2]. We randomly initialize node embedding by a uniform distribution. The dimensions of node embeddings are set to 40 for Shenzhen and 60 for Changsha, respectively. The time segment length $\Delta\tau$ is set to 30 minutes and 60 minutes. For all learning methods, we set epoch to 100, batch size to 16, and dropout to 0.3. We train our model using Adam optimizer with an initial learning rate of 0.001. Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) are adopted to evaluate the performance of each model.

$$MAE = \frac{1}{m} \sum_{i=1}^m |\hat{y}_i - y_i| \quad (23)$$

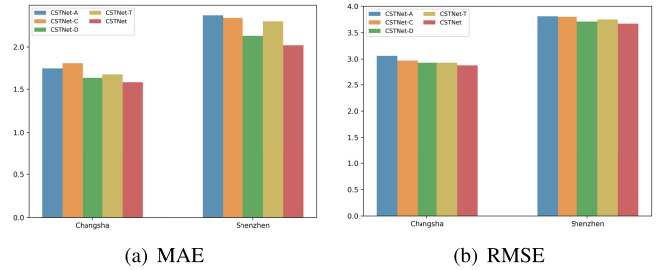


Fig. 3. Ablation study on CSTNet and four variants.

$$RMSE = \sqrt{\frac{1}{m} \sum_{i=1}^m (\hat{y}_i - y_i)^2} \quad (24)$$

C. Experimental Results

1) *Ablation Study*: The key design of our CSTNet lies in the multi-view graph structure and the interaction layer to integrate the auxiliary features with the primary feature. To validate the effectiveness of the proposed components, we compare the performance of the following four variants of CSTNet:

2) *Overall Performance*: The overall performance is presented in Table IV, where the best is remarked in bold. It is shown that the proposed CSTNet outperforms all the baselines. The performances of naive HA and SARIMA are unsatisfactory as they only consider assumptions of linear correlations over time series while overlooking the spatial dependencies among real traffic situations. The RNN-based methods, including LSTM, GRU, and OGCRNN, perform significantly better. Such results demonstrate that the deep learning-based methods can better capture the non-linear spatiotemporal correlations. The GCN-based methods, including STGCN and GWNET, outperform RNN-based methods, which benefit from the fact that the graph structure can better process and model the spatial correlations in urban road networks. Except for the

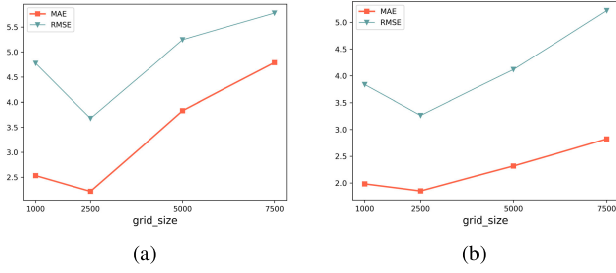


Fig. 4. Parameter Study on grid size. (a) Shenzhen. (b) Changsha.

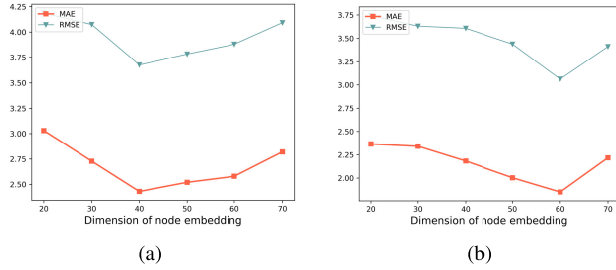


Fig. 5. Parameter Study on node embedding. (a) Shenzhen. (b) Changsha.

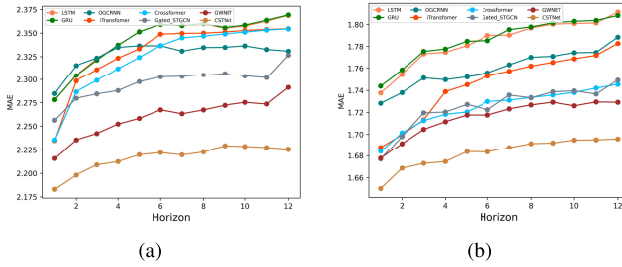


Fig. 6. Parameter Study on prediction steps. (a) Shenzhen. (b) Changsha.

proposed CSTNet, GWNEN performs the slightly best since it adopts an adaptive graph to model relationships between nodes, indicating that an adaptive graph can effectively exploit the hidden spatial dependencies.

Overall, the proposed CSTNet achieves state-of-the-art results. Particularly on Changsha dataset with time intervals of 30 minutes, the improvement of CSTNet is significant that lowers down MAE and RMSE by 11.4% and 10.4% than GWNEN. The reason is that the design of the proposed collaborative network is better suited for carbon emission prediction. Such collaborative properties stem from a multi-view graph structure of which the supplementary provides rich spatial and semantic information for the carbon emission prediction task. Beyond that, the interactions between the primary feature and auxiliary feature that share the same structure of convolution contribute a lot to this task. Via a fusion strategy, combining these two features with global factors for collaborative predicting makes the proposed CSTNet perform better than baselines constantly.

- **CSTNet-A:** It removes the whole auxiliary part and interaction layer for each ST-Conv layer.
- **CSTNet-C:** It performs the graph convolution operation using only $\mathcal{G}_c$, the spatial connectivity-based graph.
- **CSTNet-D:** It performs the graph convolution operation using only $\mathcal{G}_d$, the geographical proximity-based graph.

TABLE V
COMPARISON OF CALCULATION TIME STATISTICS

Methods	Relative Calculation Time	
	Training	Inference
HA	1.02	0.52
ARIMA	0.33	0.33
LSTM	1.96	1.97
GRU	1.94	1.98
OCGRNN	1.02	0.78
iTransformer	1.03	0.74
Crossformer	1.01	0.72
STGCN	2.51	2.52
GWNEN	2.53	2.52
CSTNet	1	1

- **CSTNet-T:** It performs the graph convolution operation using only $\mathcal{G}_t$, the geographical proximity-based graph.

Fig. 3 presents the ablation study on the CSTNet and four variants. In general, compared to variants with a single graph, the CSTNet-A without auxiliary parts performs the worst, indicating the importance of the auxiliary feature and its interaction effect. Compared to CSTNet-C with a spatial connectivity-based perspective, CSTNet-D and CSTNet-T achieve better performance. These results verify that the distance-based correlations and time similarity work better for carbon emission prediction, of which the two perspectives provide useful indication and inference.

3) *Time Complexity Analysis:* We compared the computation time of HA, ARIMA, LSTM, GRU, OCGRNN, iTransformer, Crossformer, STGCN and GWNEN with CSTNet to analyze the time complexity. The results are shown in Table V. We can observe that although CSTNet is slower than HA and ARIMA, its accuracy far exceeds these models. Compared with the single structure method, CSTNet's hierarchical structure model expresses regional traffic in the form of grids and nodes, resulting in higher training costs but correspondingly better performance. Compared to the latest methods, such as STGCN and GWNEN, CSTNet is almost twice as fast. The underlying reasons are the large number of parameters in the graph structure of STGCN and the complex network structure of GWNEN. This demonstrates the superiority of layered components with balanced overhead, and the cost of layered components is worth it. In terms of inference time, our method is not as good as some baseline methods such as iTransformer and Crossformer, but compared to training time, the proportion of inference time is smaller. Even if our method does not have an advantage in inference time, it will not fall far behind in total time. Overall, in terms of time complexity, our method is not much worse than other methods.

4) *Robustness Analysis:* In this part, we evaluate the robustness of CSTNet w.r.t. different settings of grid size, dimension of node embedding and the prediction steps. The chosen hyper-parameters are as follows: Grid size s of scheduled sampling [1000,2500,5000,7500]. The dimensions of node

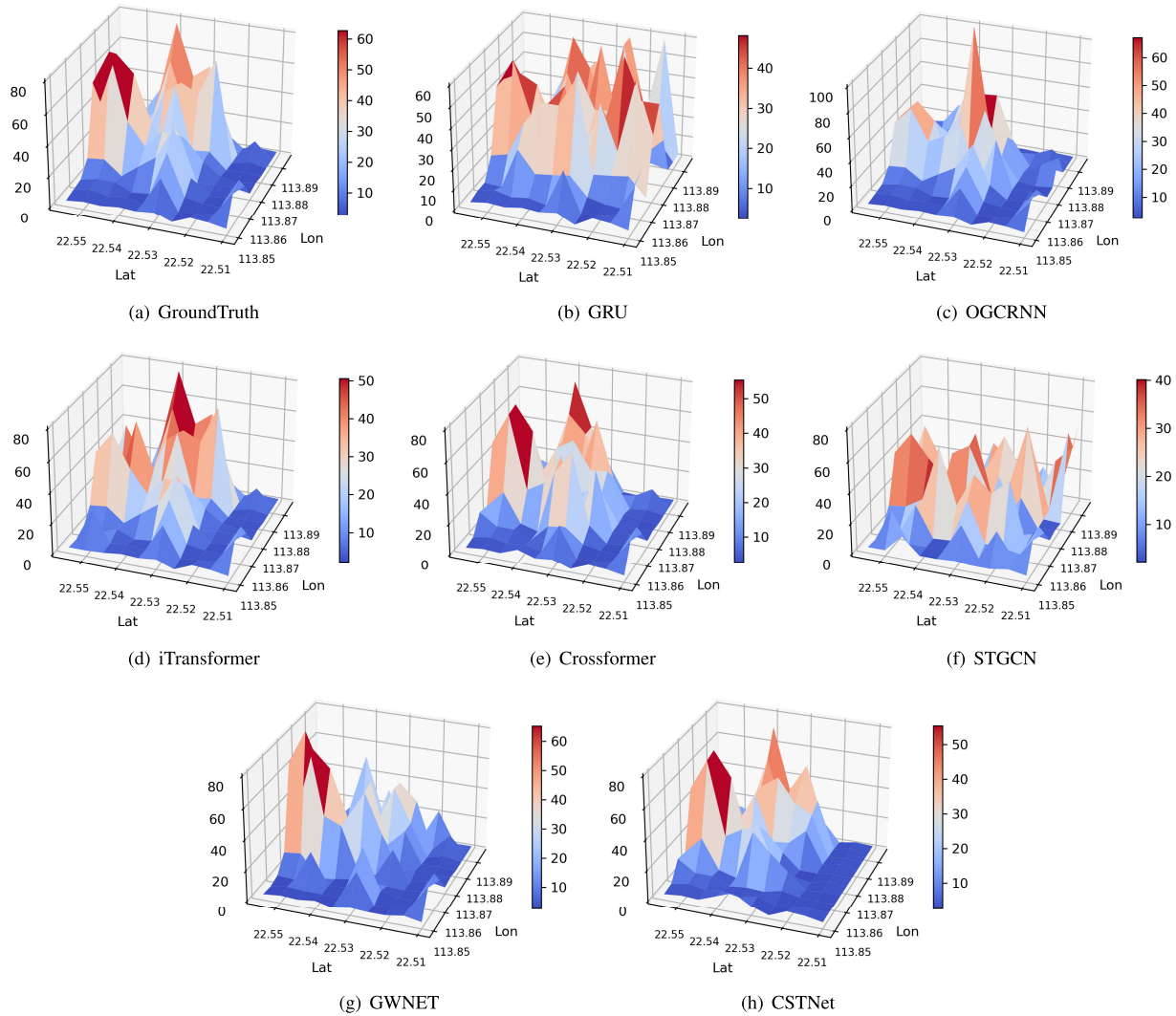


Fig. 7. Carbon emission prediction from a 3D-view.

embedding d range from 10 to 80, and we compare the results of prediction steps from 1 to 12. We repeat each experiment five times and take the average results.

Fig. 4, Fig. 5 and Fig. 6 show the analysis results on parameter sensitivity. We find that the loss of the proposed CSTNet decreases as s increases both on Changsha and Shenzhen, while keeps increasing with the growth of s from 2500 meters in Fig. 4(a) and Fig. 4(b). It is not hard to understand that if the grid is too small, the spatial correlations among trajectories would be blurred. On the contrary, if the grid is too large, the fine-grained spatial information would be lost. As illustrated in Fig. 5(a) and Fig. 5(b), we examine the dimension of node embedding for the construction of an adaptive adjacency matrix. The results show that increasing the dimension of node embedding will enhance the representation of CSTNet while reducing the MAE and RMSE.

Increasing the subsequent will degrade the representation ability of CSTNet since high dimensions will lead to over-fitting. The optimal dimension of node embedding d is 40 for Shenzhen and 60 for Changsha. It can be explained that, with the same grid size, Changsha is divided into more regions, and the traffic condition such as the varying of traffic

density is much more complicated, thus node embedding with higher dimension is required to store sufficient information. In addition, we compare the results on 12 horizons. As shown in Fig. 6 the wider the prediction step achieves, the performance of all methods gradually worsens. In this case, the proposed CSTNet remains better performance in multi-step prediction. The reason behind is that our method has great generalization ability to exploit long-interval context and hence substantially captures long-interval regularity in the carbon emission prediction.

5) Results Visualization: Furthermore, we visualize the results of proposed CSTNet and baselines to intuitively interpret the distribution of urban carbon emissions in selected 10×10 regions. Fig. 7 presents the prediction results from a 3D view. It is shown that GWNET and the proposed CSTNet outperform other baselines when compared to ground truth. Besides, Fig. 8 depicts the specific details of prediction from an overhead view.

In the following, we elaborate on the performance comparison based on the combined analysis of results. As shown in Fig. 7(a), ground truth presents the carbon emissions at the time intervals of 6:00-7:00, where the horizontal and

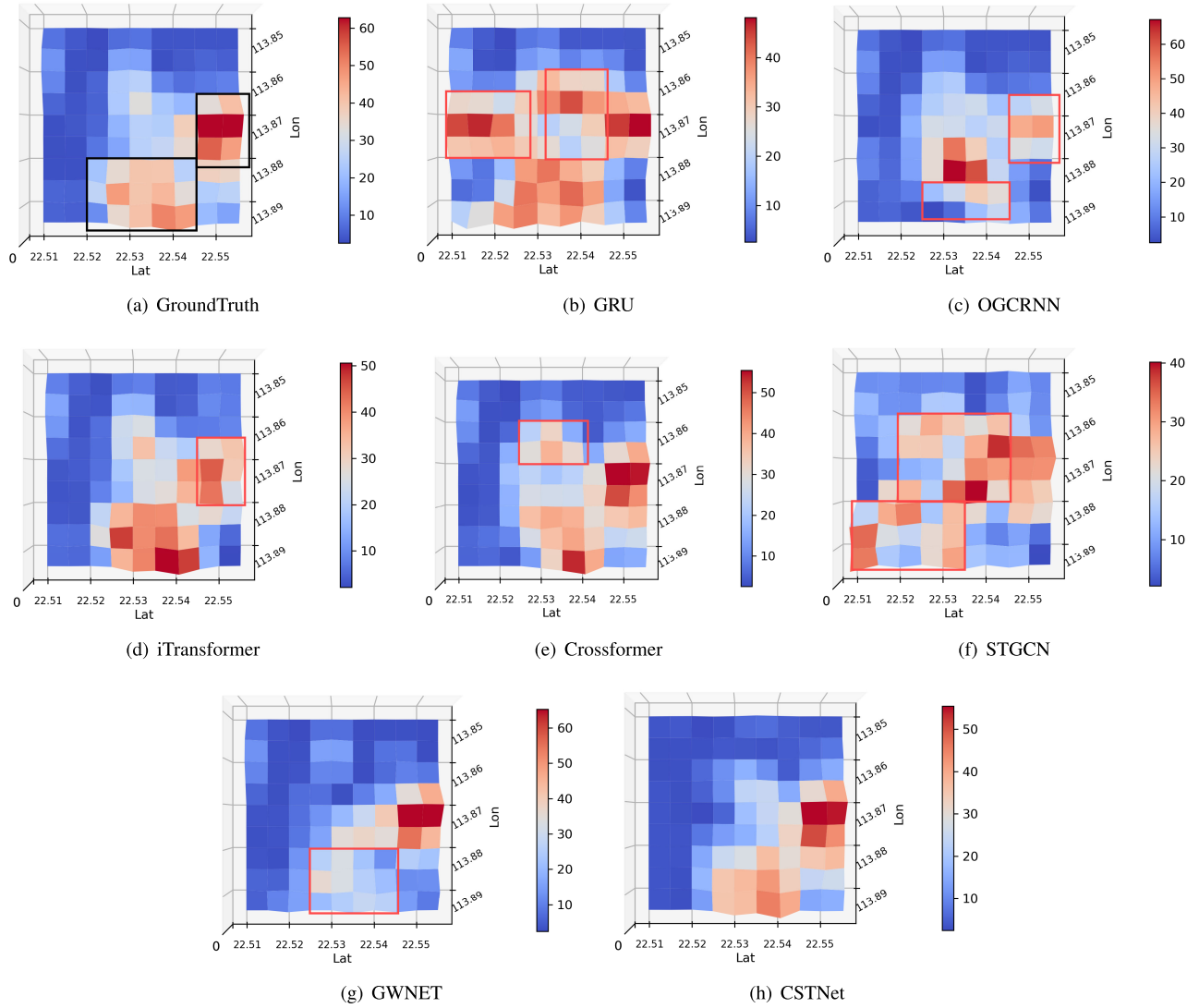


Fig. 8. Carbon emission prediction from an overhead view. The red block represents a relatively high carbon emission value, while the blue block represents the opposite.

vertical coordinates are the index number of grids. At that time, most people have not started to leave home, and fewer vehicles were traveling in the city, where almost no vehicles were in the remote regions. Accordingly, urban vehicle carbon emissions are mainly concentrated in a few grids, like residences and malls. As shown in Fig. 7(b)-7(h), readers can visually compare the predicted results of each model with the groundtruth. We can see two black rectangular are gathering peaks of carbon emissions in Fig. 8(a), as well as several potential peaks close by while most other areas are blue. As for GRU and STGCN models, they predict redundant information and can not properly capture the nonlinear variety of carbon emissions. As seen in the red rectangle of Fig. 8(b) and Fig. 8(f), they predict several incorrect carbon emission regions. The reason is that the overfitting caused by numerous data in turn brings over-predicting errors. Compared to them, OGCRNN and GWNET achieve better results, but they also produce deviations. As illustrated in Fig. 8(c) and Fig. 8(g), both have several peak carbon emission areas marked by red rectangular that are not predicted. As seen in the red rectangle

of Fig. 8(d) and Fig. 8(e), their trend predictions are very close to the groundtruth, but their predicted absolute values have a certain gap from the groundtruth. This also means that if only the approximate distribution of carbon emissions needs to be known, the prediction results of these two methods are informative. However, if used as a reference to predict whether carbon emissions in a certain area will exceed the standard in the future, this prediction result may not be satisfactory.

Overall, the shapes of peaks in the 3D view of the proposed CSTNet in Fig. 7(h) best match the real condition. Likewise, the predicted distribution of carbon emissions in Fig. 8(h) from the overhead view is closer to the true values. These results prove that CSTNet effectively captures emissions' spatio-temporal dependencies, thereby achieving satisfied prediction results.

VI. CONCLUSION

In this paper, to explore the spatio-temporal carbon emission distribution of passenger cars, we propose a collaborative spatial-temporal network (CSTNet) to predict regional carbon

emissions from passenger car trajectory data. To that end, we generate a multi-view graph structure to model multiple spatial heterogeneity. Benefiting from the constructed graph with multiple perspectives, we design a parallel structure to capture spatial-temporal dependencies for carbon emission and traffic density, respectively. Following that, we propose an interaction layer between auxiliary features and carbon emissions to handle their internal dependencies and further capture spatial relationships among them. Moreover, we identify several global factors and embed them for final prediction with a collaborative fusion. Experiments based on the large-scale passenger car trajectory dataset demonstrate that the proposed CSTNet outperforms the baselines.

Our work can help the city manager to optimize energy and environmental measures. In the future, we will devote our efforts in discovering multi-source urban data to assist in predicting carbon emissions at the region scale, such as traffic events and POIs.

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